Integer and Matrix Multiplication

Binary Long Multiplication $\Theta(n^2)$

Break into Smaller problems

e.g. 10100011×01100001

let $\begin{cases} a_1 = 1010,\ a_0 = 0011 \Rightarrow a = 2^{n/2} a_1 + a_0 \\ b_1 = 0110,\ b_0 = 0001 \Rightarrow b = 2^{n/2} b_1 + b_0 \end{cases}$

$$a \cdot b = 2^n a_1 \cdot b_1 + 2^{n/2}(a_1 b_0 + b_1 a_0) + a_0 b_0$$

```
Multiply(A, B):
n ← size of A
if n = 1 then return A[1] · B[1]
mid ← ⌊n/2⌋
U ← Multiply(A[mid+1 ... n], B[mid+1 ... n])   // a1 b1
V ← Multiply(A[mid+1 ... n], B[1 ... mid])     // a1 b0
W ← Multiply(A[1 ... mid], B[mid+1 ... n])     // a0 b1
Z ← Multiply(A[1 ... mid], B[1 ... mid])       // a0 b0
M[1 ... 2n] ← 0
M[1 ... n] ← Z   // a0 b0.
M[mid+1 ...] ← M[mid+1 ...] ⊕ V ⊕ W   // + (a1 b0 + a0 b1) · 2^(n/2)
M[2mid+1 ...] ← M[2mid+1 ...] ⊕ U   // a1 b1 · 2^n
return M
```

$\downarrow$ $\ll n/2$ bit

$T(n) = 4T(n/2) + n$ = Updating Every Bit

$= 4^i T(n/2^i) + n + 2n + 4n + \dots 2^{i-1} n$

Let $i = \log_2 n$

$T(n) = 4^{\log_2 n} + n \cdot (2^{\log_2 n} - 1)$

$= n^2 + n^2 - n = \Theta(n^2)$

Karatsuba Multiplication

$a_1 b_0 + a_0 b_1 = \underbrace{(a_1 + a_0)(b_1 + b_0)}_{\text{Extra Calculation}} - \underbrace{a_1 b_1 - a_0 b_0}_{\text{Already Calculated}}$

$\Rightarrow T(n) = 3T(n/2) + n$

$= 3^{\log_2 n} + \dfrac{1 - (\frac{3}{2})^{\log_2 n}}{1 - \frac{3}{2}} \cdot n$

$= n^{\log_2 3} + 2(1 - n^{\log_2 \frac{3}{2}}) \cdot n$

$= \Theta(n^{\log_2 3})$

Matrix Multiplication

$C_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$

Brute-force : $\Theta(n^3)$

Divide-and-Conquer

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \times \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$\begin{cases} C_{11} = A_{11} \cdot B_{11} + A_{12} \cdot B_{21} \\ \vdots \end{cases}$

8 subproblems

$T(n) = 8T(n/2) + O(n^2)$

$\Rightarrow T(n) = O(n^3)$

"[n elements]" + [n elements]

Strassen's Algorithm

$$\begin{cases} P_1 = A_{11} \times (B_{12} - B_{22}) \\ P_2 = (A_{11} + A_{12}) \times B_{22} \\ P_3 = (A_{21} + A_{22}) \times B_{11} \\ P_4 = A_{22} \times (B_{21} - B_{11}) \\ P_5 = (A_{11} + A_{22}) \cdot (B_{11} + B_{22}) \\ P_6 = (A_{12} - A_{22}) \times (B_{21} + B_{22}) \\ P_7 = (A_{11} - A_{21}) \times (B_{11} + B_{12}) \end{cases} \Rightarrow \begin{cases} C_{11} = P_5 + P_4 - P_2 + P_6 \\ C_{12} = P_1 + P_2 \\ C_{21} = P_3 + P_4 \\ C_{22} = P_5 + P_1 - P_3 - P_7 \end{cases}$$

$T(n) = 7T(n/2) + n^2$

$= T(n^{\log_2 7})$

Master Theorem

$T(n) = aT(n/b) + f(n)$

$a \geq 1$, $b > 1$ ★

$f(n)$: asymtotically positive polynomial function ($O(n^2) \to n^2$)

let $c = \log_b a$

① if $f(n) = O(n^{c-\epsilon})$, $\epsilon > 0$

$\Rightarrow T(n) = \Theta(n^c) = \Theta(n^{\log_b a})$

② if $f(n) = \Theta(n^c)$

$\Rightarrow T(n) = \Theta(n^c \log n)$

③ if $f(n) = \Omega(n^{c+\epsilon})$

$\Rightarrow T(n) = \Theta(f(n))$